PHY203 - Mechanics Lab

Bachelor, Semester 3
Ecole Polytechnique
By: Allan Lizarraga and Mayad Mnif

The crazy cable car

In this lab project we will study the motion of a toy model of a cable car (Fig. 1 left), consisting in a cart rolling on a track and carrying a rigid pendulum (Fig. 1 right). From the measurement of the positions of the cart and pendulum, we will show that all coordinates are not independent of each other as they follow so-called "holonomic constraints". We will also infer the kinetic and potential energies, demonstrate transfers between those and study energy conservation. Finally, using magnets, we will tweak the potential energy and observe the changes in the dynamics of the system.

Figure 1: A cable car (left) and the toy model under study (right).

The experiments will require you to take movies with the GoPro camera that is supplied and to analyze these movies using the TracX Jupyter notebook. You will then export position data and visualize / compute quantities using python via the provided Jupyter notebook.

1 Holonomic constraints

From the previous acquisitions, we have the coordinates of the cart (X_c, Y_c) and of the pendulum (X_p, Y_p) as a function of time, all in the reference frame of the lab. We will show that these 4 coordinates are not independent.

1.1. First, plot Y_c as a function of X_c. Is the knowledge of X_c sufficient to know unambiguously the value of Y_c?

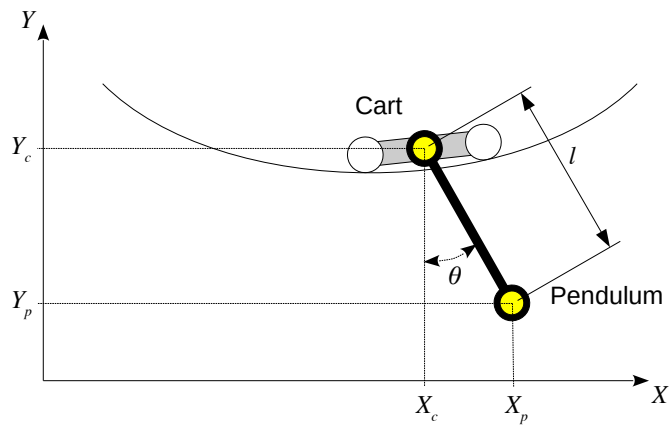
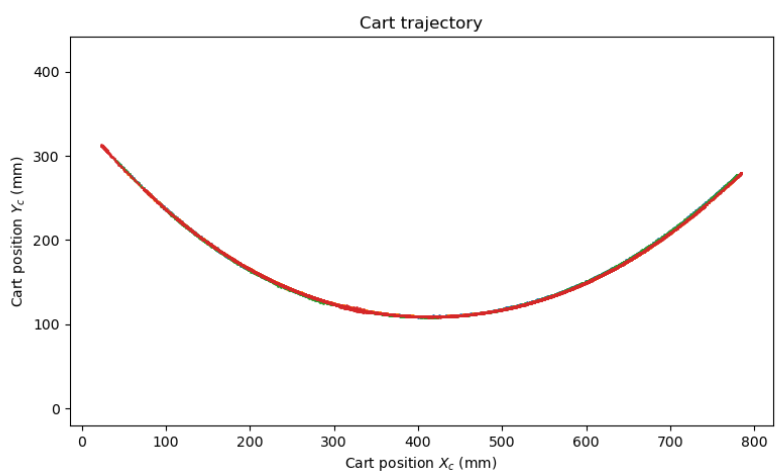


Figure 2: Sketch of the setup and definition of the coordinates.



Solution:
The graph shows the cart's trajectory, and we can see that it follows the curve of the function $Y_c = f(X_c)$. Therefore with every X_c we can find the corresponding Y_c.

1.2. Could you approximate the relationship $Y_c = f(X_c)$ without tracking the cart at all (note: no explicit mathematical expression of f is expected)?

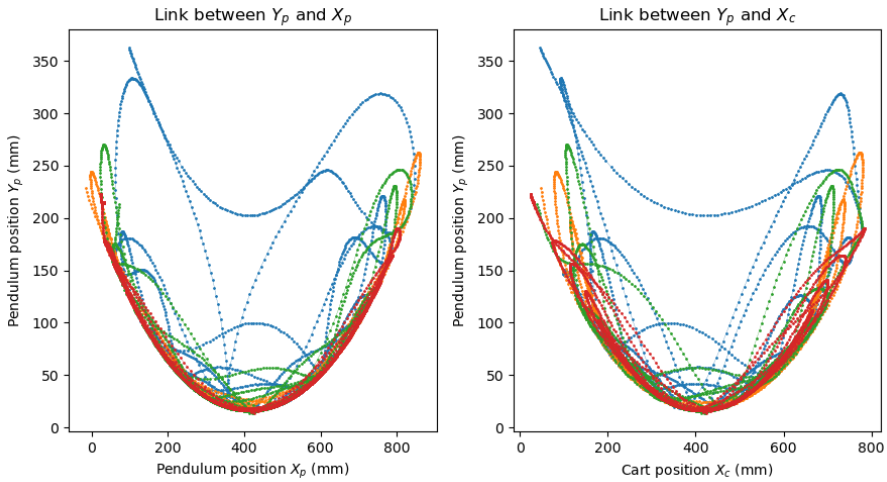
Solution: Yes, since the cart is constrained to the track, Y_c is a single-valued function of X_c defined by the path that the track takes. We could determine it using the TrackX script and fitting a collection of (X,Y) points along the path with a polynomial.

1.3. Is the relation valid for any motion of the cart?

Solution: Yes because the cart is always constrained to the track, thus Y_c will always depend on X_c.

1.4. Now plot Y_p as a function of X_c or X_p. Is there a simple relation between Y_p and one of the two horizontal coordinates?

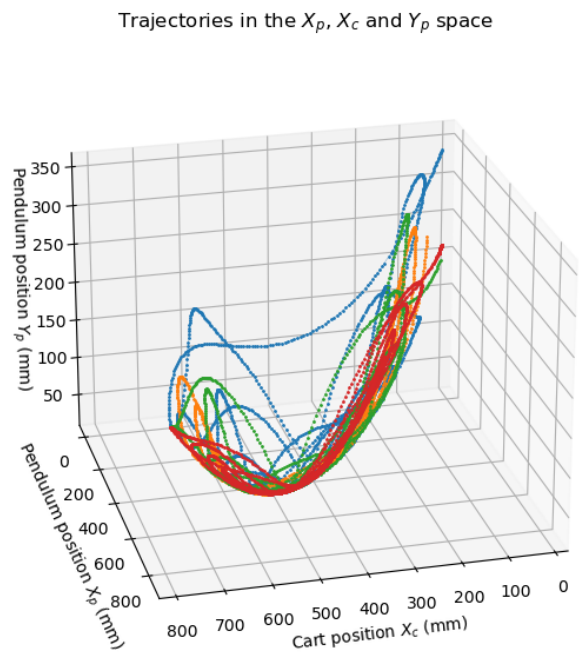
Solution:



The two plots show the evolution of the pendulum's vertical position Y_p as a function of the pendulum position X_p (left) and the cart position X_c (right). Each color corresponds to a different initial condition, so a different starting angle of the pendulum. The red curve represents the case where the pendulum was released without any initial angle, the other curves correspond to pendulums initially displaced to one side before release.

From the plots we can see that there is no simple relation between Y_p and either X_c or X_p. The vertical position of the pendulum depends not only on the horizontal position of the cart but also on the angle and motion of the pendulum relative to it. Because these quantities change dynamically and depend on the initial conditions, Y_p cannot be expressed as a function of X_c or X_p.

1.5. Using a 3D plot, plot Y_p as a function of X_c **and** X_p. Describe the shape of the surface that these points trace.

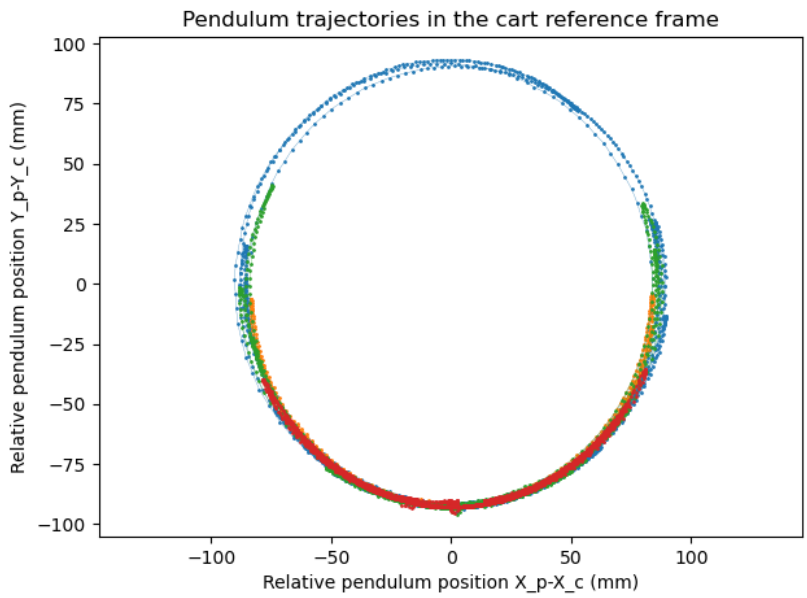


Solution:
The 3D plot shows the trajectories of the pendulum in the (X_c, X_p, Y_p) space for different initial conditions. Each colored curve corresponds to a different initial angle of the pendulum.

We observe that all trajectories start close to the same region but then diverge depending on the pendulum's angular motion. The curves form a "banana" shaped surface (which we could later on better observe in other groups' graphs) that follows the parabola of the track.

The surface traced by the points is due to the coupled motion between the cart and the pendulum. Near the bottom of the track, Y_p follows the vertical trajectory of the cart, but as the pendulum swings, Y_p deviates from this trajectory. The surface therefore represents the sum of the cart's path and the pendulum's oscillations, producing a curve where the height Y_p varies with both X_c and X_p.

1.6. These measurements indicate that there exists a relation that can be written as $h_2(X_c, X_p, Y_p) = 0$ (no exact analytic expression for h_2 is expected). What is the geometrical meaning of this constraint?



Solution: This equation represents a physical constraint that we have in our system. From the analysis of the previous question, it was found that the pendulum goes around the cart with a circular constraint of radius l, the length of the pendulum. Geometrically, this means that the motion of the pendulum is not free in 3D: its coordinates (X_c, X_p, Y_p) are limited to a specific orbit determined by the circular constraint of the pendulum lenght. The way h_2 can be zero from a mathematical point of view is by having the interaction of the different coordinates give us a hard constraint, which as stated before is the trajectory of the pendulum. The constraint $h_2 = 0$ is visible on the graph plotting the pendulum trajectories in the cart reference frame. The trajectories describe a clear circle, of radius l.

In general, constraints between coordinates X_i (with $i = 1$ to N) are called holonomic if they can be written in the form $h_j(X_1, \ldots, X_N) = 0$ (with $j = 1$ to $J \leq N$). Such relations reduce the number of degrees of freedom from N to $N - J$. Here you have shown that the $N = 4$ coordinates can actually be reduced to 2 degrees of freedom thanks to 2 holonomic constraints:

$$h_1(X_c, Y_c) = Y_c - f(X_c) = 0$$

$$h_2(X_c, X_p, Y_p) = 0$$

We can therefore choose 2 generalized coordinates to describe the system, for example X_G (horizontal position of the center of gravity of the system) and θ (the angular position of the pendulum).

1.7. Write all 4 initial coordinates as a function of the 2 generalized coordinates.

Solution:

The system has four coordinates, but the two constraints reduce it to two degrees of freedom. We choose as generalized coordinates:

X_G : horizontal position of the center of mass

θ : angle of the pendulum with the vertical
We have

$$X_G = \frac{m_c X_c + m_p X_p}{m_c + m_p}$$

From the geometry of the pendulum, the pendulum position can be written relative to the cart:
$X_p = X_c + l \sin\theta$
$Y_p = Y_c - l \cos\theta$
Substituting $X_p = X_c + l \sin\theta$ into X_G, the equation gives:

$$X_G = \frac{m_c X_c + m_p (X_c + l \sin\theta)}{m_c + m_p}$$

Simplifying:

$$X_G = X_c + \frac{m_p l \sin\theta}{m_c + m_p}$$

Rearranging for X_c:

$$X_c = X_G - \frac{m_p l \sin\theta}{m_c + m_p}$$

Then,

$$X_p = X_c + l \sin\theta = X_G + \frac{m_c l \sin\theta}{m_c + m_p}$$

Since the cart is constrained to the track, its vertical position is a function of its horizontal coordinate:

$$Y_c = f(X_c)$$

where $f(X_c)$ represents the track.
The pendulum's vertical coordinate then follows from geometry:

$$Y_p = Y_c - l \cos\theta$$

We obtain all four initial coordinates as functions of the two generalized coordinates X_G and θ:

$$X_c = X_G - \frac{m_p l \sin\theta}{m_c + m_p}$$

$$Y_c = f(X_c) = f(X_G - \frac{m_p l \sin\theta}{m_c + m_p})$$

$$X_p = X_G + \frac{m_c l \sin\theta}{m_c + m_p}$$

$$Y_p = Y_c - l \cos\theta = f(X_c) = f(X_G - \frac{m_p l \sin\theta}{m_c + m_p}) - l \cos\theta$$

These relations show how the motion of the system is fully determined by the horizontal position of the center of mass X_G and the pendulum angle θ.

Mechanical systems can also be subject to non-holonomic constraints, such as those that involve the speed or when they involve inequalities instead of equalities. Picture for example what would happen if the rigid rods of the pendulum were replaced by strings.

1.8. Can you imagine such a non-holonomic constraint for some movements in the two systems (string/rigid rod)?

> **Solution:** If we wanted to analyze a similar system without holonomic constraints, we could replace the rigid rod of the pendulum with a string. This change would transform h_2 into an inequality instead of an equality, because the string can only exert tension and not compression.
>
> Another example is about the track: if we were to replace the smooth track with a rough one, the system would experience frictional forces that depend on the direction and speed of the motion. Then, the constraint would no longer be only geometric thus becoming a non-holonomic constraint.

2 Energy conservation and fluxes

From the previous acquisitions, one can compute the speed $(\dot{X}_c, \dot{Y}_c)$ of the cart and $(\dot{X}_p, \dot{Y}_p)$ of the pendulum using a discrete time derivative in the

processing software. The Jupyter notebook computes these two speeds from the trajectories you feed it, and then computes the kinetic T, potential V, and mechanical energy E of both masses ($m_c = 150\ g$ and $m_p = 50\ g$).

Energy	**Kinetic**	**Potential**	**Mechanical**
Cart	$T_c = \frac{1}{2}\ m_c\ (\dot{X}_c^2 + \dot{Y}_c^2)$	$V_c = m_c\ g\ Y_c$	$E_c = T_c + V_c$
Pendulum	$T_p = \frac{1}{2}\ m_p\ (\dot{X}_p^2 + \dot{Y}_p^2)$	$V_p = m_p\ g\ Y_p$	$E_p = T_p + V_p$
System	$T = T_c + T_p$	$V = V_c + V_p$	$E = T + V = E_c + E_p$

2.9. Plot the time evolution of the various energies as a function of time. Observe that energy can transfer from one term to the other during the course of an experiment, such that any of the terms can increase or decrease. Annotate the graphs with you observations to support them.

> **Solution:**
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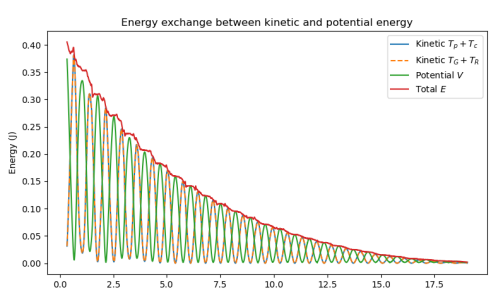
> 
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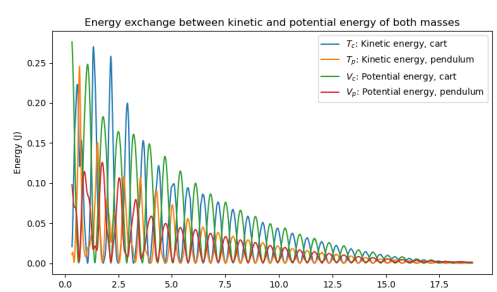
> 
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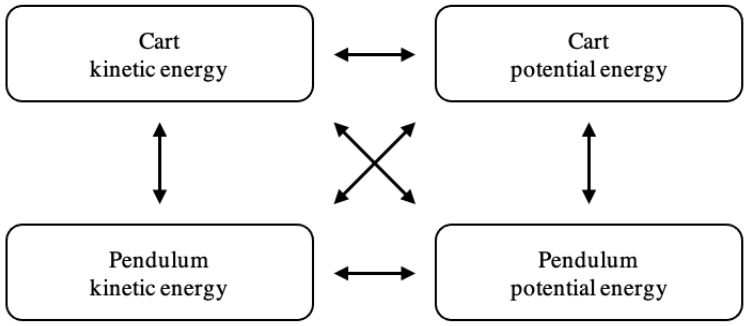


From the first graph it is possible to observe how the total kinetic and potential energies change over time, when one increases, the other decreases at the same rate with a gradual energy loss that we should expect from a physical system. Now in the second graph the energies between the cart and the pendulum seem chaotic and stabilize slowly over time. In the first (chaotic) stage we have a coupled system between the cart and the pendulum, where one injects energy to the other and vice versa, this is due to the initial conditions where if the pendulum has a higher initial potential energy, we will have more of a chaotic behavior. In the second stage we have what appears like a single system acting, this happens when the system stabilizes and we obtain the classic oscillation between kinetic and potential energy from the cart and the pendulum working in tandem.

Now looking at a time-evolution of the different terms:

2.10. Does the total energy of the system increase?

Solution: As we can observe in both graphs of the previous question, total energy decreases over time. This tells us that there is an energy loss in the system which in part can be from the friction from the cart rolling in the track and the components connecting the pendulum to the cart. The sudden drops observed on the curve for the total energy is due to the friction. Indeed, at kinetic energy spikes, the velocity of the cart is at a maximum, and therefore the friction force is also at a maximum, resulting in a loss of energy. There are also spikes that indicate that energy is "increasing" in some local points, this is not physically possible without external forces acting on our system, so we can cross of that deduction since our experiment didn't have external forces interacting. The only possible cause to this are the numerical approximations made by the python code to solve the energies when we have a fluctuation.

2.11. Can you identify periods in the time series that are dominated by (1) the kinetic energy of the cart, (2) the potential energy of the cart, (3) the potential energy of the pendulum, (4) the kinetic energy of the pendulum? You may need to acquire new trajectories with *ad hoc* initial condition to answer this question. What limits the contribution of each term to the total energy?

Solution:

From the energy time series, we can identify four distinct regimes corresponding to different dominant energy terms:

1. When the cart moves freely with little pendulum motion, the kinetic energy of the cart T_c dominates. This occurs when the cart undergoes significant horizontal motion while the pendulum remains near its equilibrium position. This is limited by the fact that the car can't go at a greater kinetic energy that the initial potential energy, this is also diminished as time goes by because of energy loss

2. When the cart changes height along the track, the potential energy of the cart U_c becomes dominant. This happens near the highest or lowest points of the track, where the cart's elevation Y_c changes substantially. This is limited by the initial conditions, the cart can't reach a higher position on the track than the one it was originally on.

3. When the pendulum reaches its highest position, the potential energy of the pendulum U_p is maximum. At this instant, the pendulum's velocity is nearly zero and its height Y_p is maximal. This is limited by the fact that it slowly losses energy and because of it's lower mass, eventually it depends completely on the carts movement.

4. When the pendulum passes through its lowest position, the kinetic energy of the pendulum T_p peaks, since the pendulum's velocity is largest at the equilibrium position. This is limited by the mass it has and the fact that over time it will eventually stay in the equilibrium position.

2.12. Is the total energy conserved? Why?

Solution: In an idealized system without any form of dissipation, the total mechanical energy would remain constant over time. However, experimental data shows a gradual decrease of $E(t)$, indicating that the total energy is not perfectly conserved.

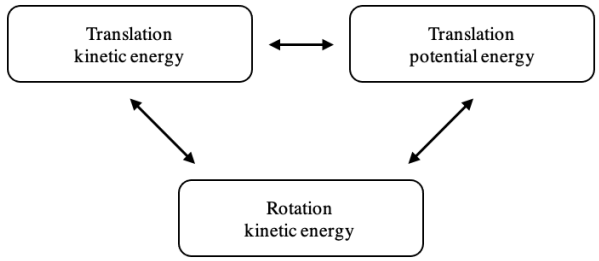
This loss of mechanical energy can be attributed to non-conservative forces such as friction between the cart and the track, air resistance acting on the moving parts, internal damping at the pendulum hinge.

These effects convert a fraction of the system's mechanical energy into heat and sound, resulting in a slow but continuous decline of the total measured energy over time.

Switch to the generalized coordinates X_G and θ and compute again the kinetic, potential and mechanical energies. Even though these two variables are sufficient to describe the state of the system at any moment, explicitly writing the energies on this basis requires writing a relationship between X_G, Y_G, and θ. This is possible but will lead to messy mathematical formulas. For this reason, we will keep all three variables and use them as measured experimentally.

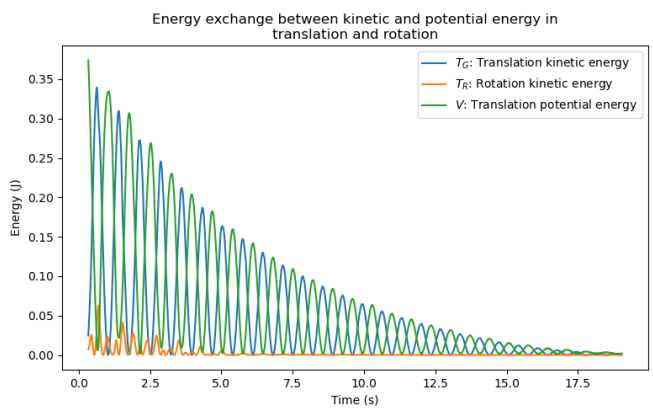
Energy	**Kinetic**	**Potential**	**Mechanical**
Translation	$T_T = \frac{1}{2}\ m\ (\dot{X}_G^2 + \dot{Y}_G^2)$	$V_T = m\ g\ Y_G$	$E_T = T_T + V_T$
Rotation	$T_R = \frac{1}{2}\ I\ \dot{\theta}^2$		$E_R = T_R$
System	$T = T_T + T_R$	$V = V_T$	$E = T + V = E_T + E_R$

with $m = m_c + m_p$, $I = m_c\ m_p\ l^2\ /\ (m_c + m_p)$, and $l = 87\ mm$ the pendulum length.



2.13. Observe one more time the fluxes between the kinetic and potential energies, and comment on the role of damping.

Solution:



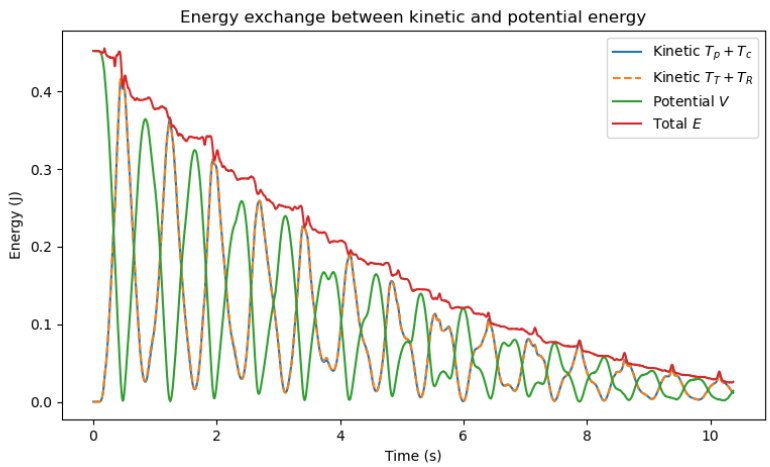
From the viewpoint of generalized coordinates we can observe how rotational kinetic energy is only significant in the first few seconds introducing some changes into the translational energy while the system stabilizes. Translational energy maintains an almost steady fluctuation between kinetic an potential energy signalizing the different positions of our center of gravity on the track. It is clear that there is a damping effect that eventually takes energy to 0 due to the conditions of the experiment which is exactly what we would expect from a real physical system without external forces.

3 Potential energy profile

A strong magnet is embedded on the bottom face of the cart. Using a second magnet that you will place below the track, you can add a new potential field in addition to gravity, either attractive or repulsive. Be aware that the magnetic field has a short range of interaction but a strong amplitude, so take precaution when choosing the initial conditions to change the dynamics of the cart in a controlled fashion. Perform new experiments and record new data with both signs of interaction.

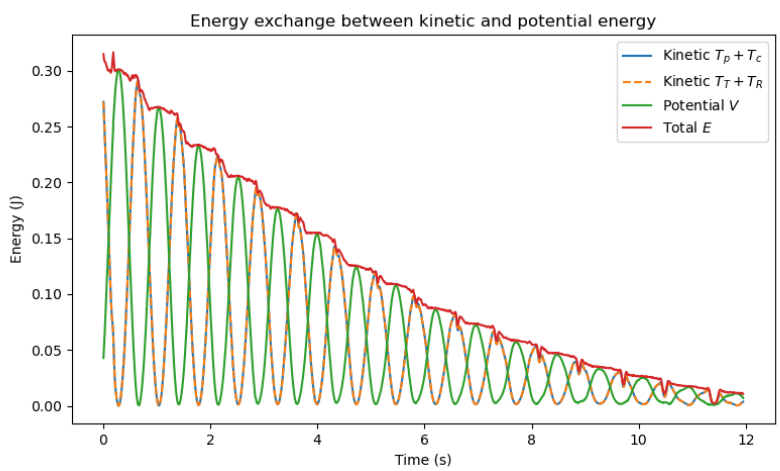
3.14. Plot the kinetic, potential and total mechanical energy of the system, as a function of time and/or cart position. Do you observe any difference in the total energy? Can you explain it?

Solution: We consider the cases of an attractive and repulsive magnet.
Attractive case:



Again, we find a decreasing mechanical energy of the system. However we can see the effects of the magnet on the energy exchange, after a few initial oscillations where the energy of the magnetic field is negligible compared to the energy of the cart + pendulum system. At low potential energy (at the bottom of the ramp) the magnet interacts with the cart. When the cart approaches the magnet, it is accelerated towards it and gains kinetic energy, which is again lost when escaping the magnet. This results in spikes in the mechanical energy, because the magnet introduces the energy to the closed system.

Repulsive case:



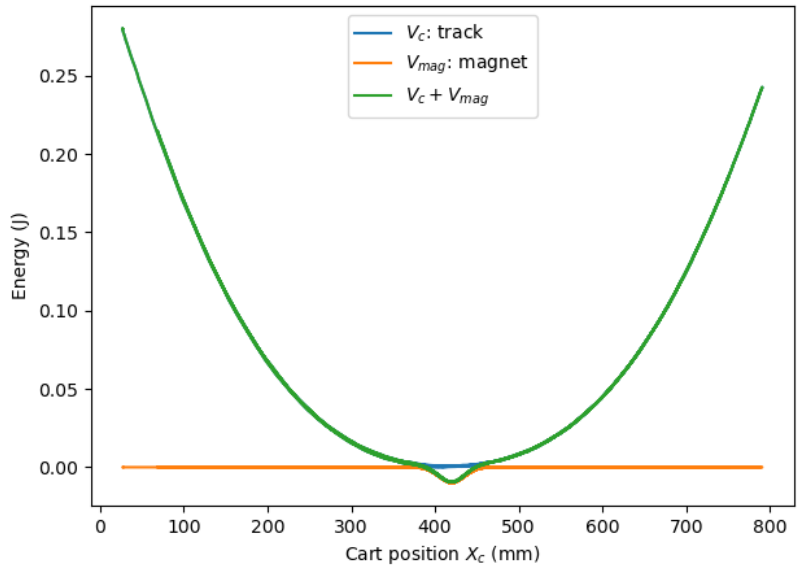
In the repulsive case, we have similar observations, but this time we see dips and not spikes in the total energy of the system, since the magnet decelerates and then accelerates the cart (contrary to the previous case).

3.15. Now that the potential energy is not simply given by the altitude of the track, can you estimate and plot the new potential energy profile $V_c(X_c)$, by comparing these traces with the data obtained without a magnet?

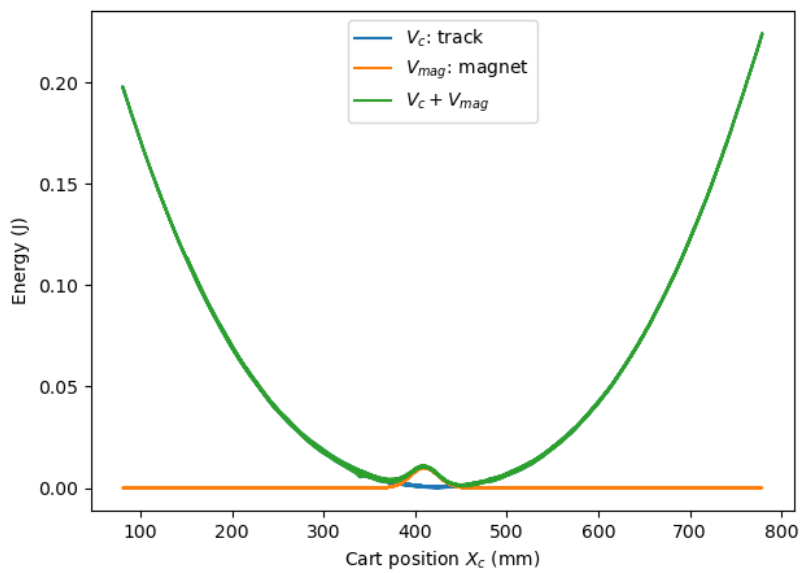
Solution: By superposing the potential of the magnet and the gravitational potential, we obtain the total potential. We know the gravitational potential is mgY_c, but we cannot find analitically

the solution for V_{mag}. To model V_{mag}, we use a Gaussian distribution, for which we derive the parameters experimentally. We can then plot the new potential energy profile, as the sum of the two potentials V_c and V_{mag}, to compare it with the data obtained without magnet.

Attractive case:



Repulsive case:

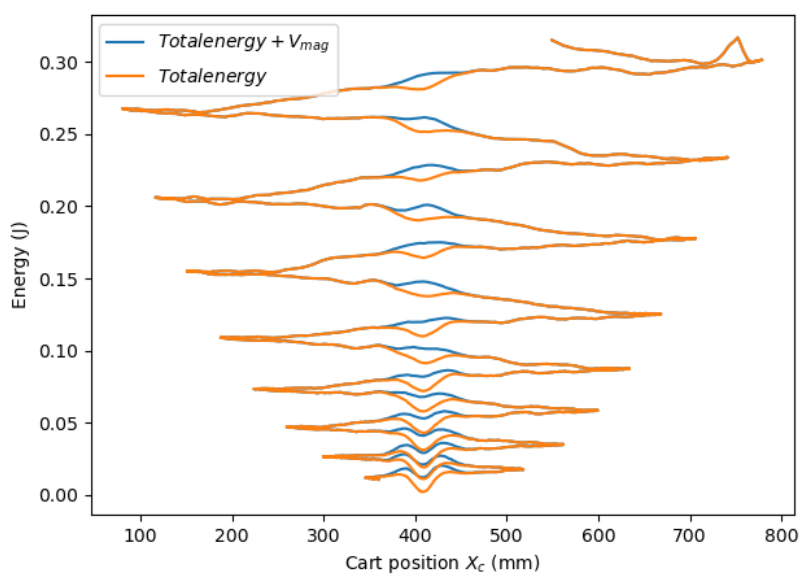


In both graphs we notice that Vmag dominates the potential energy profile of the system the cart is near the bottom of the ramp, hence regions where $Vc \propto Yc \approx 0$. The magnet represents a hill in the repulsive case or a dip in the attractive case for the cart to pass. In the attractive case, there is one stable equilibrium where the cart stops, centered on the magnet. In the case of the repulsive magnets we see two points of stable equilibrium, equidistant to the center point the ramp,

hence explaining why the cart will not come to rest at the center of the ramp, but to the side of the magnet.

3.16. How high or deep is the bump added on the track due to the magnet? How wide is it? Trick: we do not know the analytical shape of the energy profile, but from the experimental observation we can assume that it has a limited range. A reasonable approach is to use a Gaussian curve: $V_{mag}(X_c) = m\ g\ h_{mag}\ e^{(-(X_c-X_0)^2/2\ \Delta X^2)}$, where h_{mag} is the equivalent bump height or depth according to its sign, X_0 its horizontal position, and ΔX its width.

Solution: To determine the dimensions of the bump (focusing on the repulsive case) due to the magnet, we use the Gaussian curve for the magnetic potential. The goal is to tweak the parameters of the Gaussian curve until the curve for the total energy ($V_{mag} + V_c$) loses the spikes of energy near $X_c = 0$.



Mechanical energy as a function of X_c (repulsive magnet)

This is the final plot we obtained using the parameters $V_{mag,max} = 0.01J$ and $\Delta X = 18mm$. From this data, we can directly compute the dimensions of the bump.

$$V_{mag,max} = mgh_{mag}$$

We thus easily get the height of the bump:

$$h_{mag} = \frac{V_{mag,max}}{mg} = \frac{0.01J}{0.2kg \times 9.81m.s^{-2}} = 5mm$$

In the repulsive case, we find that the magnet at the bottom of the track can represent a bump of height $5mm$ and of width $18mm$. For the case of the attractive magnet, the sign is simply inverted, so the bump would be $5mm$ deep.

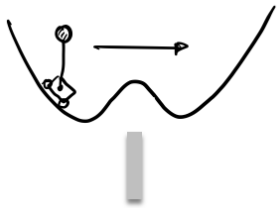
4 Bonus experiments: challenges

This is a bonus section, just for fun, do not worry if you do not have time to test all chalenges. No need to use Tracker to analyze the videos, you may just describe your protocol in your report. Make sure to show your videos or live demonstrations to your instructors to validate your challenges!

4.17. **Cross the barrier**: in the repulsive configuration, can you find a proper static initial condition (no kinetic energy) to cross the potential energy barrier with the cart initially below this barrier?

Solution: When the pendulum is below the magnet's level, the cart cannot cross. When the pendulum is at or above the magnet's level, the cart can cross. This means that the total mechanical energy must exceed the magnetic potential energy barrier.

4.18. **Avoid an obstacle**: in the repulsive configuration, can you find a proper initial condition to cross the potential energy barrier and avoid (at least once) an obstacle (a piece of paper) on the pendulum trajectory?



Solution: When the pendulum's center of mass is positioned above the cart and slightly in the direction of motion, it remains balanced above the cart as the system moves forward. This is because the inertial forces acting on the pendulum help maintain this equilibrium during motion. With this configuration, the cart can cross the magnetic potential energy barrier while the pendulum stays elevated, so that it can avoid the obstacle on its trajectory at least once.

4.19. **Loop**: with no initial kinetic energy and no magnet, what is the maximum number of full turns you can observe for the pendulum? How would you tune the track and pendulum to maximize or minimize this number of turns?

Solution: When performing the experiment, the maximum numbers of turns we could get was 4, by placing the cart at the highest point on the track to have ass much potential energy as possible, which is then converted in rotational kinetic energy for the pendulum. To maximize the number of turns we could make the track higher, so that the cart can have more potential energy and thus more mechanical energy throughout the motion. Also, making the pendulum longer would allow the pendulum to have a much higher moment of inertia ($I \propto l^2$), and therefore more rotational energy, resulting in a higher number of turns. To minimize the number of turns, we simply lower the track and reduce the length of the pendulum. Moreover, increasing the friction of the track or the damping of the pendulum's hinge would result in bigger energy losses during the motion which would decrease the rotational energy of the pendulum.